COMMON KNEE JOINT PROBLEMS

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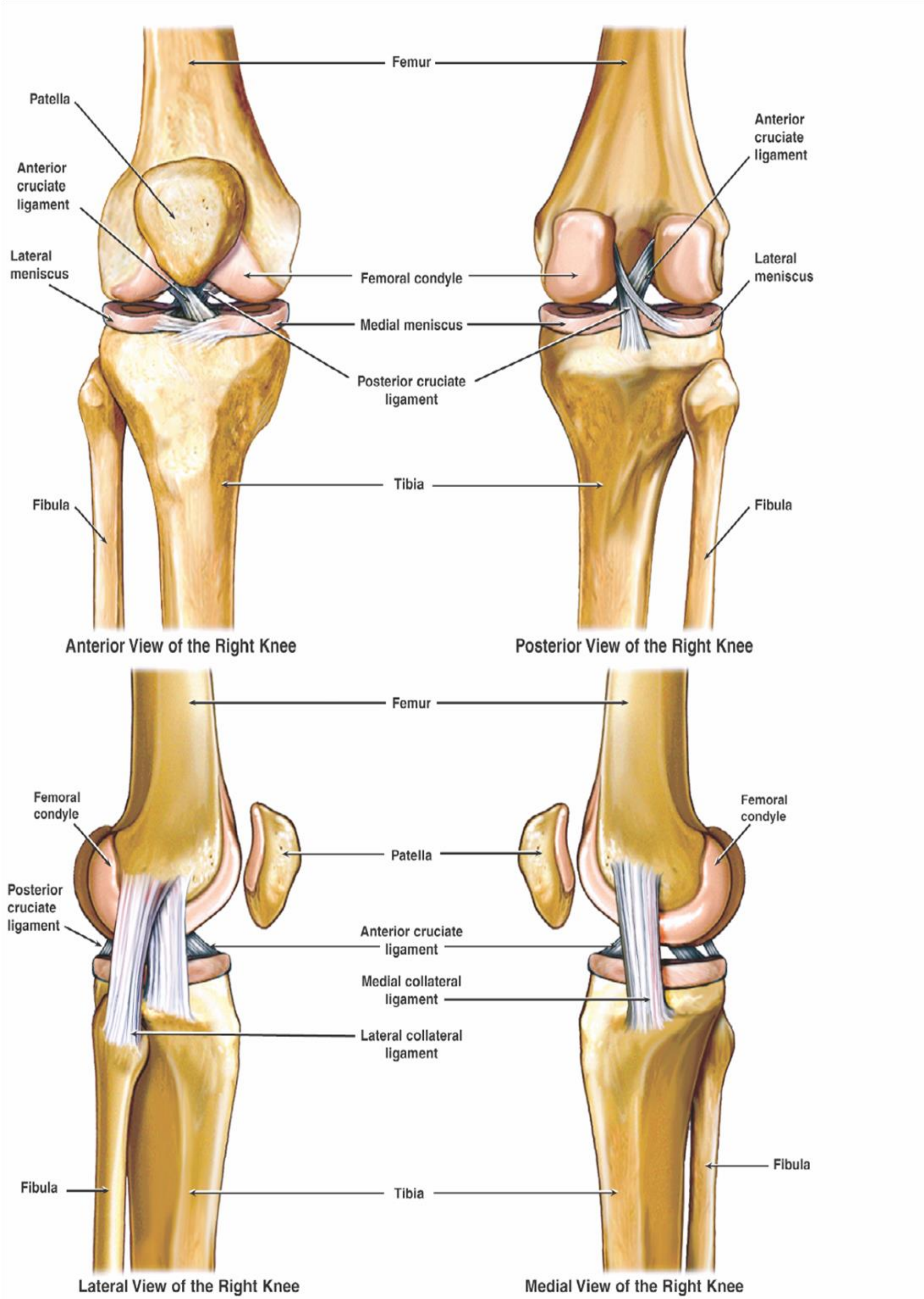
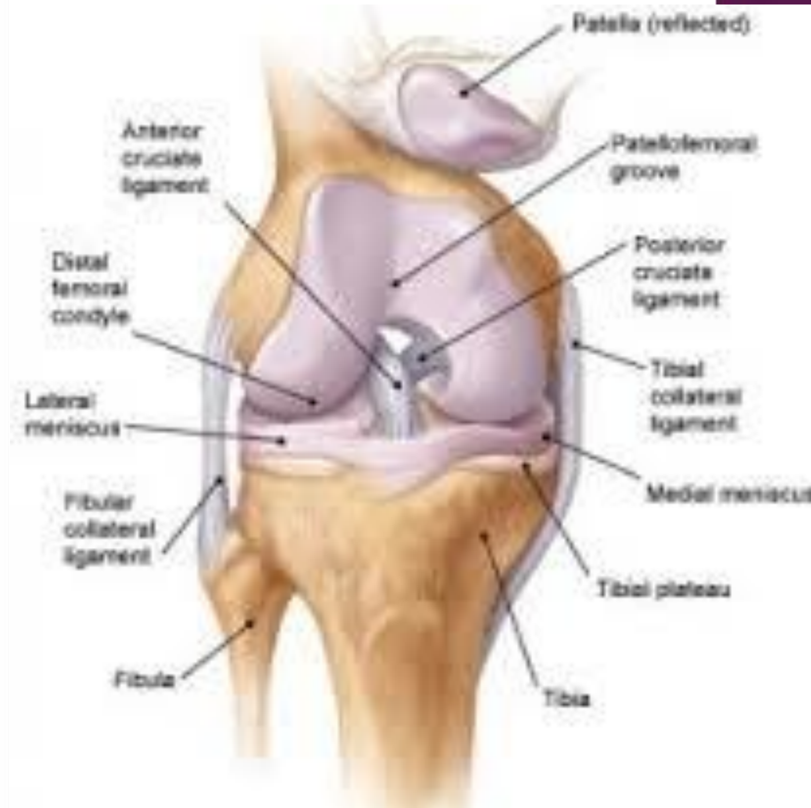
ANATOMY

- Composed of articulation between the distal femoral condyle and tibial plateau and bet.the femoral condyle and patella (sesamoid bone)
- It is a hinge joint
- Synovial joint
- Because of the shape of the bone it depend on the ligaments as a major stabilizing element

WHAT ARE IMPORTANT LIGAMENTS

- ACL
- PCL
- LCL
- MCL


Femur
Patella
Anterior cruciate ligament
Lateral meniscus
Femoral condyle
Medial meniscus
Posterior cruciate ligament
Tibia
Fibula
Anterior cruciate ligament
Lateral meniscus
Fibula
Anterior View of the Right Knee
Posterior View of the Right Knee
Femur
Femoral condyle
Patella
Femoral condyle
Posterior cruciate ligament
Anterior cruciate ligament
Medial collateral ligament
Lateral collateral ligament
Fibula
Tibia
Fibula
Lateral View of the Right Knee
Medial View of the Right Knee


Patella (reflected)
Anterior cruciate ligament
Patellofemoral groove
Distal femoral condyle
Posterior cruciate ligament
Lateral meniscus
Tibial collateral ligament
Fibular collateral ligament
Medial meniscus
Tibial plateau
Fibula
Tibia

KNEE PROBLEMS

- History of trauma ,previous Hx of trauma ,pain,catching ,clicking, locking
- Examination
- Investigation
- Blood investigation CBC&ESR
- FBS
- S.uric acid

IMAGING

- Pain X-ray
- CT scan
- MRI
- Bone scan techniscium
- Arthrography

IMAGING OF THE KNEE

- X-ray we need AP,Lateral ,axial view,intercondylar notch or tunnel view
- Ap view should be in standing position
- CT scan used to detect #
- MRI …for ligamentous injuey
- Radioscitigraphy…detect secondaries occult infection in joint replacement


Base of patella
Patella
Anterior surface
Apex of patella
Tibia
Tibial tuberosity
Intercondylar fossa
Lateral condyle
Apex of head


Shaft of femur, Body of femur
Base of patella
Patella
Apex of patella
Lateral epicondyle
Intercondylar fossa
Groove for popliteus
Lateral condyle
Lateral intercondylar tubercle
Anterior intercondylar area
Lateral condyle
Apex of head
Head
Neck
Tibial tuberosity
Fibula
Adductor tubercle
Medial epicondyle
Medial condyle
Medial condyle
Tibia

CLINICAL FEATURES

- The knee is swollen ,tense ,painfull ,skin is bruised some time gap can be felt
- Active extension should be tested if the Pt can extend the knee it means that the extensor mechanism is intact

PATELLAR DISLOCATION

- Because of the normal valgus aliengment of the knee there is a tendency of the patella to dislocate laterallywhen thequaricps muscle contract
- Mechanism of injury
- 1. indirect While the knee is flexed and quads

are contracted dislocation will occur

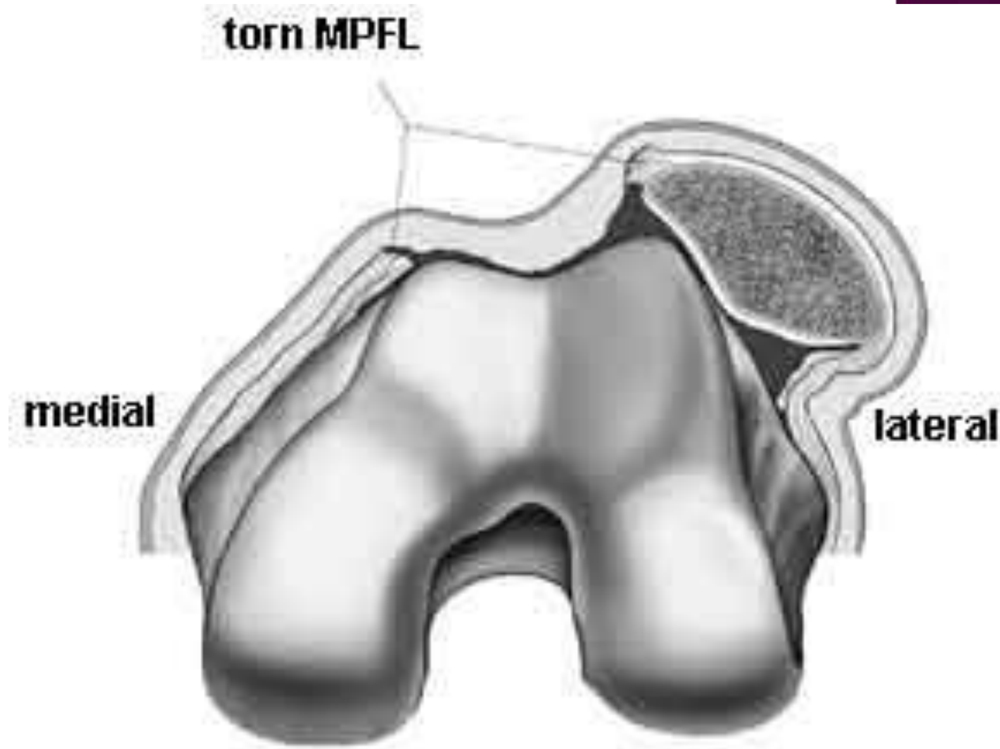
2. Direct direct force applied to the lateral part while the knee forced to valgus and external rotation

CLINICAL FEATURE

- There is tearing sensation and falling on the ground most of the times the patella rturns back to its position and sometimes remains dislocated on the lateral side of the knee as prominent lump
- Rarely there may be intraarticular dislocation to the intercondylat notch

IMAGING

- Xray shows the classical dislocation and sometimes there may be osteochondral fracture


torn MPFL
medial
lateral

TREATMENT

Most of the cases can be treated conservatively by replacing the patella to its position with or without anesthesia for 2-3 weeks S.T surgery required ror ruptured medial patellofemoral liamgamrnt

KNEE DISLOCATION

- Knee dislocation happen after considerable force by RTA or FFH it is associated with rupture of the ACL,PCL,LCL,MCL with or without #
- Clinical features
- There is hemarthrosis with bruising and soft tisssue laceration loss of the normal shape of the knee there is 40% chances of injury to the popliteal vessel.

- And there is 20% injury to the common peroneal nerve
- X-ray shows classical dislocation ST there may be avulsion of the tibial spine or collateral ligaments or # of the head of the fibula
- ST artiography my be needed in cses of vascula injry
- MRI shows the pattern of ligament injury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TREATMENT

- Immediate reduction under anesthesia avoiding hyperextension in order not to tension the poplitael vessel the splint is applied and checking of the circulation done repeatedly for 48 hr
- Vascular injury need urgent intervension with application of Ex.fixation
- Early reconstruction of all the ligament is done when the patient become stable by arthroscopic technique

ACUTE INJURY TO THE EXTENSOR APPARATUS

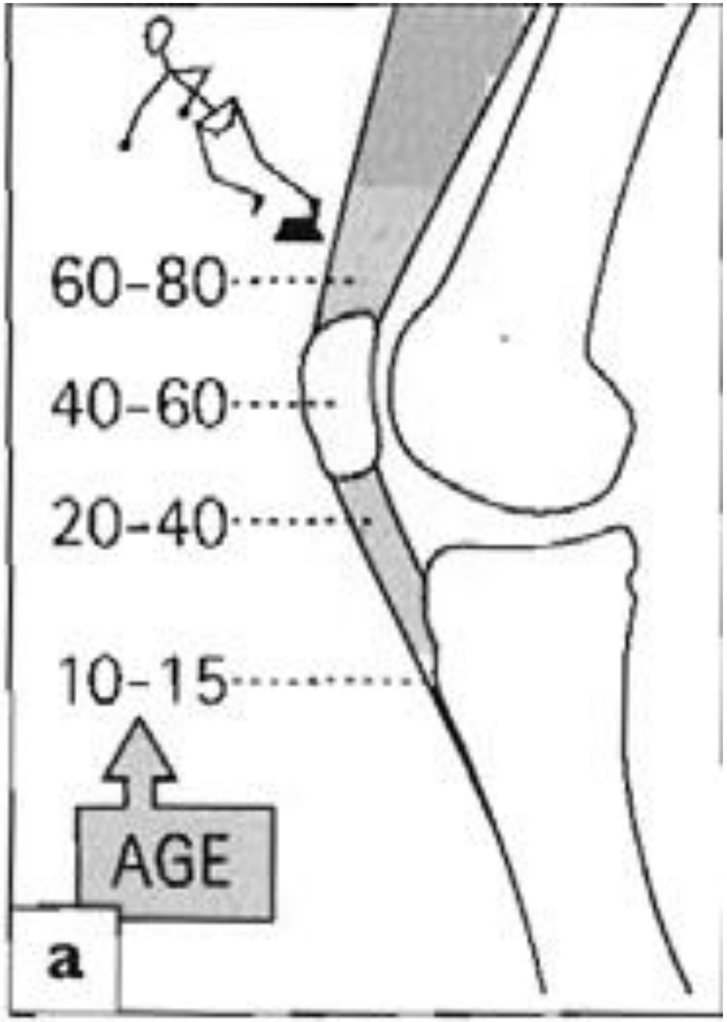
- Disruption of the extensor apparatus occurs at the following sites
- 1.avulsion of the tibial tubercle at adolescent
- 2.young adult …..rupture of the patellar tendon
- 3. middle aged …..# patella
- 4.older people and those with chronic illness….rupture of the quariceps tendon

RUPUTURE OF THE PATELLAR TENDON

- This uncommon injuey happen in young athletes ,the tear may in the proximal or distal attachment of the ligaments there may be previous Hx of local steroid injection to the ligament
- C.F: Hx of sudden sever pain and swelling on forced extension
- X-ray showed high riding patella or klake of bone from the proximal or distal part of tye ligament

DIAGNOSTIC CALENDAR

- Congenital disorder …present either at birth or in the 20^{th} or 30 year
- Adolescent…anterior knee pain due to patellar instability,plica syndrome,or osteochondritis
- Young adult with sport activities …meniscal injury or ligamentous injury
- Above midddle age…mostly degenerative changes OA either primary or secondary


60-80
40-60
20-40
10-15
AGE
a

DEFORMITIES OF THE KNEE

- By the end of the growth the knee will be in 5-7 degree of valgus anything from this regarded as abnormal
- Bow legs and knock knees in children

- distance between the knees with the child standing
- and the heels touching; it should be less than 6 cm.
- Similarly, knock knee can be estimated by measuring
- the distance between the medial malleoli when the
- knees are touching with the patellae facing forwards;
- it is usually less than 8 cm.

PHYSIOLOGICAL BOW LEGS AND KNOCK KNEES

- Bow legs in babies and knock knees in 4-year-olds are so common that they are considered
- to be *normal stages of development*
- Other postural abnormalities
- such as ‘pigeon toes’ and flat feet may coexist
- the parents should be reassured and the child should be seen at intervals of 6 months to record progress.

- In the occasional case where, by the age of 10, the deformity is still marked (i.e. the intercondylar distance is more than 6 cm or the intermalleolar distance more than 8 cm), operative correction should be advised.

PATHOLOGICAL BOW LEG AND KNOCK KNEE

- Disorders which cause distorted epiphyseal and/or physeal growth may give rise to bow leg or knock knee; these include some of the skeletal dysplasias and the various types of rickets, as well as injuries of the epiphyseal and physeal growth cartilage.

GENU RECURVATUM (HYPEREXTENSIONOF THE KNEE)

- *Congenital recurvatum This may be due to abnormal* intra-uterine posture; it usually recovers spontaneously.
- Hereditory like generalized joint laxity
- Inflammatory like RA
- Paralytic condition like poliomyelitis
- Trauma like growth plate injuries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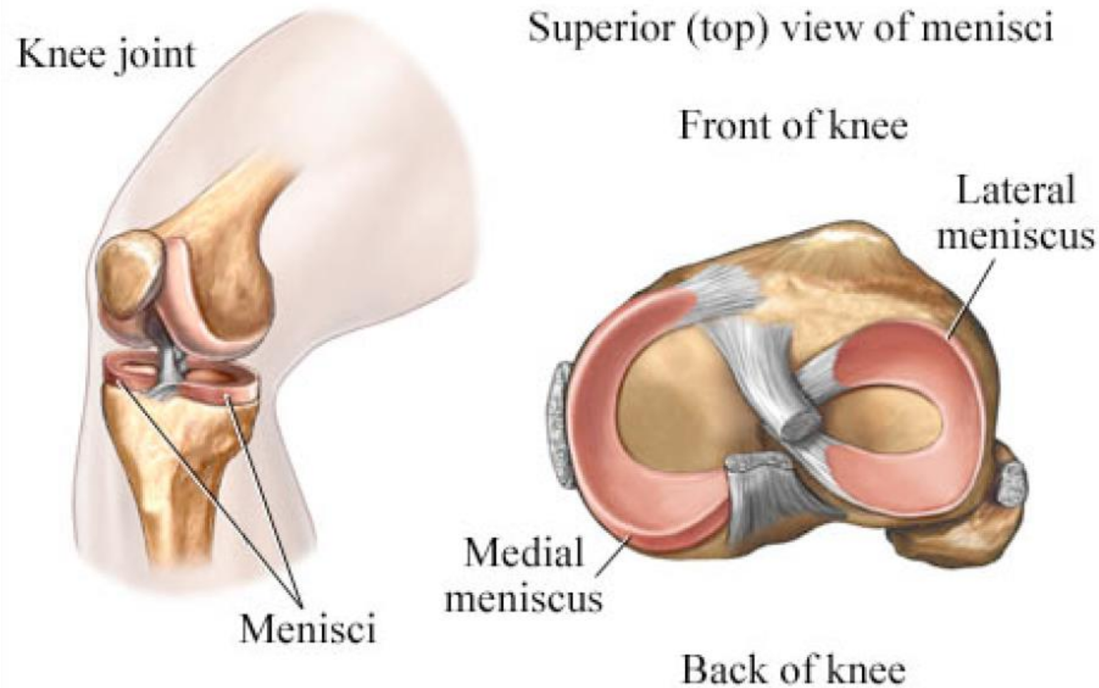
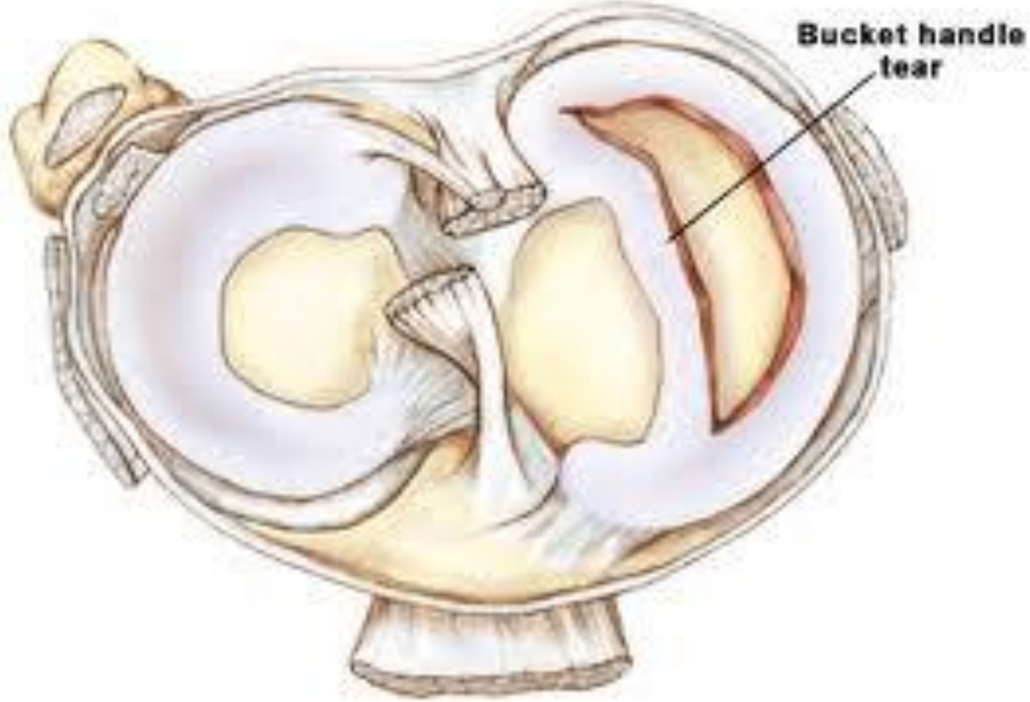
LESIONS OF THE MENISCI

- The menisci have an important role in
- (1) improving articular congruency and increasing the stability of the knee,
- (2) controlling the complex rolling and gliding
- actions of the joint and
- (3) distributing load duringmovement.

- If the menisci are removed, articular stresses are markedly increases.
- The medial meniscus is much less mobile than the lateral, and it cannot as easily to accommodate to abnormal stresses.
- This may be why meniscal lesions are
- more common on the medial side than on the lateral

Knee Joint
Cartilages (Menisci)




Bucket handle
tear

- Even in the absence of injury, there is gradual stiffening and degeneration of the menisci with age, so
- splits and tears are more likely in later life – particularly if there is any associated arthritis

TEARS OF THE MENISCUS

- The split is usually initiated by a rotational grinding force, which occurs (for example) when the knee is flexed and twisted while taking weight;
- hence the frequency in footballers.
- In middle life, when fibrosis has restricted mobility of the meniscus, tears occur with
- relatively little force

- Most of the meniscus is avascular and spontaneous repair does not occur unless the tear is in the outer third, which is vascularized from the attached synovium and capsule.
- The loose tag acts as a mechanical irritant, giving rise to recurrent synovial effusion and,
- in some cases, secondary osteoarthritis

CLINICAL FEATURES

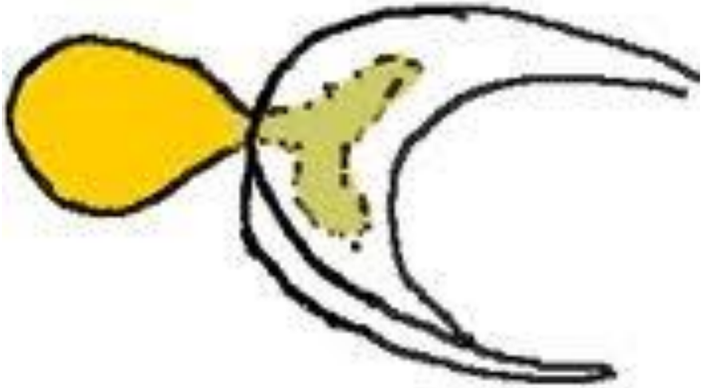
- Pain (usually on the medial side) is often severe and further activity is avoided; occasionally the knee is ‘locked’ in partial flexion.
- Almost invariably, swelling appear s some hours later, or perhaps the following day.
- Sometimes the knee gives way spontaneously and this again followed by pain and swelling.
- Locking’ – that is, the sudden inability to extendthe knee fully – suggests a bucket-handle tear

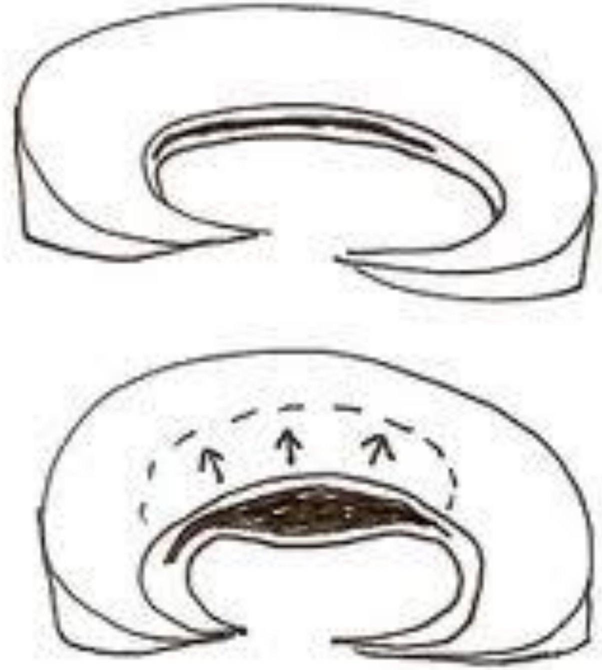
INVESTIGATION

- Plain x-ray
- MRI
- Arthroscopy
- Treatment :acute phace rest in knee splint in extension ,daily physiotherapy ,ice packs application for 3-4 weeks in hope that thee tear will heal
- Opeative : if the knee is locked and can not be reduced ,frequent giving way , then arthroscopic menisectomy or repair done

MENISCAL CYSTS

- Cysts of the menisci are probably traumatic in origin, arising from either a small horizontal cleavage tear or repeated squashing of the peripheral part of the meniscus.
- It is also suggested that synovial cells infiltrate into the vascular area between meniscus and capsule and there multiply.





CLINICAL FEATURE

- Lump on the sides of the joints slightly below joint line
- Intermittent pain esp.after activity
- Treatment : if the cyst is symptomatic arthroscopic decompression with partial menisectomy done

OSTEOCHONDRITIS DISSECANS

- An increase in the OCD has been observed in recent years, probably due to the growing participation of young children of both genders in competitive sports.
- A small, well-demarcated, avascular fragment of bone and overlying cartilage sometimes separates from one of the femoral condyles and appears as a loose body in the joint.

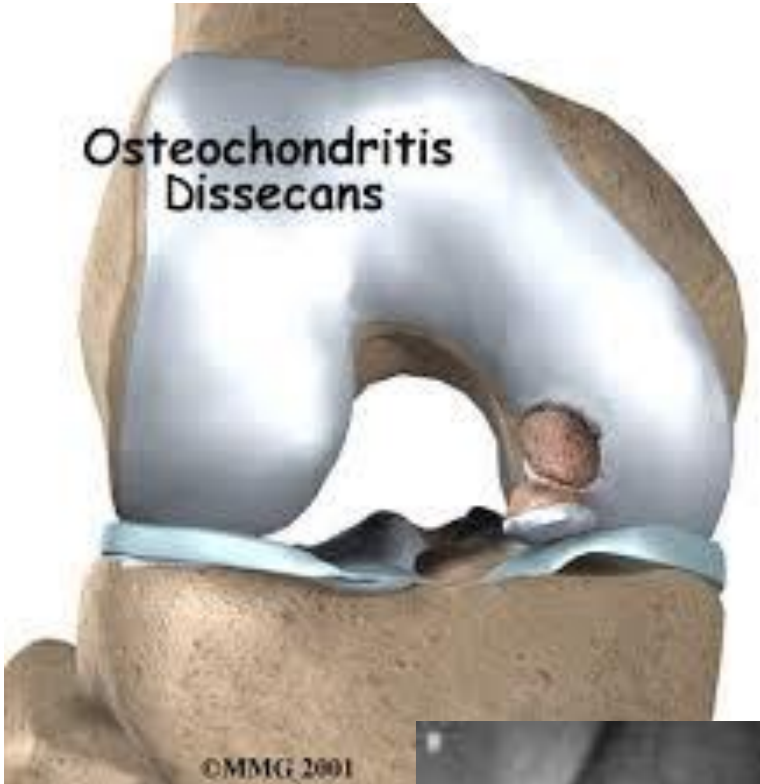
- The most common cause is trauma
- In 80 % of the cases the site is the medial aspect of the lateral femoral condyle condyle
- **Pathology**
- An area of subchondral bone becomes avascular and within this area an ovoid osteocartilaginous segment is demarcated from the surrounding bone.
- At first the overlying cartilage is intact and the fragment is stable

over a period of months the fragment separates but remains in position;

- finally the fragment breaks free to become a
- loose body in the joint. The small crater is slowly filled with fibrocartilage, leaving a depression on the articular surface

- **Clinical features**
- The patient, usually a male aged 15–20 years, presents with intermittent ache or swelling. Later, there are attacks of giving way such that the knee feels unreliable‘locking’ sometimes occurs
- The quadriceps muscle is wasted and there may be a small effusion. Soon after an attack there are two
- signs that are almost diagnostic.

- : (1) tenderness localized to one femoral condyle; and
- (2) Wilson’s sign: if the knee is flexed to 90 degrees, rotated medially and then gradually straightened, pain is felt; repeating the
- test with the knee rotated laterally is painless


Osteochondritis
Dissecans
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TREATMENT

- Those lesions with an intact articular surface have the greatest potential to heal with non-operative treatment if repetitive impact loading is avoided.
- *In the earliest stage, when the cartilage is intact and* the lesion is 'stable', no treatment is needed but activities are curtailed for 6–12 months. Small lesions often heal spontaneously

- ***If the fragment is 'unstable'***, *i.e. surrounded by a* clear boundary with radiographic 'sclerosis' of the underlying bone, or showing MRI features of separation,
- treatment will depend on the size of the lesion.
- A small fragment should be removed by arthroscopy and the base drilled; the bed will eventually be covered by fibrocartilage, leaving only a small defect.
- large fragment (say more than 1 cm in diameter) should be fixed in situ with pins

- ***If the fragment is completely detached*** *but in one piece* the crater is cleaned and the floor drilled before replacing the loose fragment
- and fixing it with special screws
- In recent years attempts have been made to fill the residual defects by articular cartilage transplantation: either the insertion of osteochondral plugs harvested from another part of the knee or the application of
- sheets of cultured chondrocytes.

Figure 6 – Arthroscopic view of treatment of a subchondral OCD lesion by means of microfracturing; arrows indicate the limits of the lesion.

LOOSE BODIES

- may be produced by:
- (1) injury (a chip of bone orcartilage);
- (2) osteochondritis dissecans (which may
- produce one or two fragments);
- (3) osteoarthritis(pieces of cartilage or osteophyte);
- (4) Charcot’s disease(large osteocartilaginous bodies).
- (5) synovialchondromatosis (cartilage metaplasia in the
- synovium,

CLINICAL FEATURES

- Loose bodies may be symptomless. The usual complaint is attacks of sudden locking without injuryThe joint gets stuck in a position which v aries from one attack to another
- A pedunculated loose body may be felt; one that is truly loose tends to slip away during palpation (the well-named ‘joint mouse’).

TREATMENT

- A loose body causing symptoms should be removed unless the joint is severely osteo arthritic. This can usuallybe done through the arthroscope.

OSTEOARTHRITIS

- The knee is the commonest of the large joints to be affected by osteoarthritis .
- Often there is a predisposing factor:
- injury to the articular surface,
- torn meniscus,
- ligamentous instability
- preexisting deformity of the hip or knee.
- in many cases no obvious cause can be found

- Osteoarthritis is often bilateral and in these cases there is a strong association with Heberden’s nodes and generalized osteo arthritis

- changes are most marked in the medial compartment.
- The characteristic features of cartilage fibrillation,sclerosis of the subchondral bone and peripheral osteophyte formation are usually present;
- in advanced cases the articular surface may be denuded of cartilage and underlying bone may eventually crumble

- Clinical features
- Patients are usually over 50 years old; they tend to be overweight and may have long standing bow-leg deformity.
- Pain is the leading symptom, worse after use, or (if the patello-femoral joint is affected) on stairs.
- Afterrest, the joint feels stiff and it hurts to ‘get going’ after sitting for any length of time. Swelling is common,
- giving way or locking may occur.

- On examination there may be an obvious deformity (usually varus) or the scar of a previous operation.
- The quadriceps muscle is usually wasted

Movement is somewhat limited and is often

- accompanied by patello-femoral crepitus
- The natural history of osteoarthritis is one of alternating ‘bad spells’ and ‘good spells

- X-ray
- The anteroposterior x-ray *must be obtained with the* patient standing and bearing weight; only in this way can small degrees of articular cartilage thinning be revealed. The tibio-femoral joint space is diminished
- (often only in one compartment) and there is subchondral sclerosis. Osteophytes and subchondral cysts are usually present and sometimes there is soft-tissue calcification in the suprapatellar region or in the joint itself (chondrocalcinosis).
- .

- Treatment
- If symptoms are not severe, treatment is conservative.
- Joint loading is lessened by using a walking stick. Quadriceps exercises are important. Analgesics are prescribed for pain, and warmth (e.g. radiant heat or shortwave diathermy) is soothing.
- Intra-articular corticosteroid injections will often relieve pain,

- New forms of medication have been introduced
- in recent years, particularly the oral administration of glucosamine and intra-articular injection of hyalourans.
- There is, as yet, no agreement about the
- long-term efficacy of these products.

OPERATIVE TREATMENT

- *Arthroscopic washouts, with trimming of degenerate* meniscal tissue and osteophytes
- *Realignment osteotomy is often successful in relieving* symptoms and staving off the need for 'end-stage' surgery.
- *Replacement arthroplasty is indicated in older*

  patients with progressive joint destruction.

CHARCOT'S DISEASE

- Charcot's disease (neuropathic arthritis) is a rare cause of joint destruction. Because of loss of pain sensibility and proprioception, the articular surface breaks down and the underlying bone crumbles. Fragments of
- bone and cartilage are deposited in the hypertrophic synovium and may grow into large masses

Clinical features

The patient chiefly complains of instability; pain (other than tabetic lightning pains) is unusual. The joint is swollen and often grossly deformed

SWELLINGS OF THE KNEE

1.swelling of the entire joint
2. swellings in front ofthe joint
3. swellings behind the joint;
4. bony swellings.

ACUTE SWELLING

- 1.acute swelling
- A.traumatic haemarthrosis
- Swelling immediately after injury means blood in the joint. The knee is very painful and it feels warm, tense and tender. Later there may be a ‘doughy’ feel. Movements are restricted. X-rays are essential to see if there is a fracture; if there is not, then suspect a tear of the anterior cruciate ligament

- 2.bleeding disorder like haemophilia ,christmas disease
- 3.septic arthritis the joint is hot tender and painfull and tense ,the infecting organism usually is staphylococcus and in adult gonococcal infection should be excluded
- Joint aspiration reveal purulent discharge
- ESR and WBC are highly elevated
- Treatment is urgent if the pus is thin repeated aspiration with appropriate antibiotics given by i.v route if the pus is thick or if there is no response after 36 hr surgical drianage should be done

- Traumatic synovitis
- Chronic swelling
- 1.arthritis like osteoarthritis and rheumatoid arthritis
- 2. synovial disorder like synovial chondromatosis and pugmented villonodular synovitis and T.B arthritis